Messaging Security & Privacy Report: 24 Chat Services Compared

Report date: 2 October 2026 · Data current to 2 October 2026

This report compares the security and privacy of WhatsApp, Facebook Messenger, Telegram (cloud chats and secret chats scored separately), Signal and MEGA Chat, plus Viber, Arattai, Botim, X Chat, Google Chat, Nostr direct messages, Google Messages (RCS) and Filen Chats, the direct-message features of Discord, Steam, Reddit, LinkedIn and Instagram and the workplace and business tools Microsoft Teams, Slack, Zoho Cliq, Zoom Team Chat and WhatsApp Business, as of October 2026. It examines encryption design, metadata collection, data sharing, backups, openness, server locations and legal jurisdiction, and the 2024–2026 incident record, then scores each on eight weighted criteria. Signal ranks first at 9.3/10; Telegram cloud chats rank last at 3.4/10.

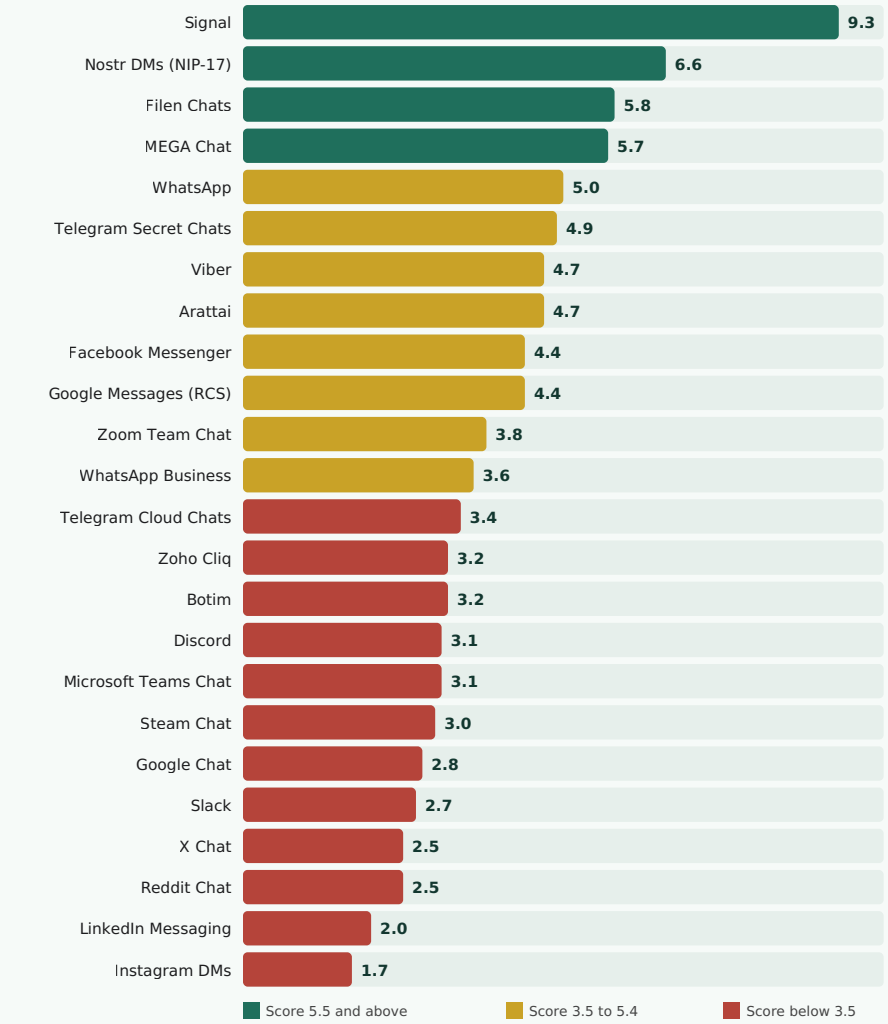
At a glance

Which chat app can the operator read?

Security & privacy scores for 24 messaging services · weighted 0–10 · October 2026

10 of 24	9 of 24	1	1
encrypt text E2EE by default	encrypt 1:1 calls by default	dropped encryption in 2026	scores above 7/10

Weighted score (/10), ranked



Four things the scores say

1. **The cipher is not the difference.**
   WhatsApp and Signal use the same protocol for content; the 4-point gap is metadata, ads and who runs the server.
2. **Calls are better protected than text.**
   Nine services encrypt 1:1 calls by default; only ten do the same for messages, and work tools make it optional.
3. **Default beats opt-in.**
   Telegram secret chats and Instagram's old opt-in mode show that encryption few people switch on protects almost no one.
4. **A licence or a label is not an audit.**
   Botim, X Chat, Zoho Cliq resellers and Steam all claim encryption that no published protocol or audit backs up.

www.wealthyhealthy.pro Messaging Security & Privacy Report · sources and method inside

Executive summary

Signal is the only one of the twenty-four configurations that scores above 6/10 on security and privacy; Instagram DMs score lowest since Meta removed their encryption in May 2026. Scores are weighted across eight criteria (method in the next section) and reflect the public record as of 2 October 2026.

Rank	Service	Weighted score (/10)	One-line verdict
1	Signal	9.3	Default E2EE with post-quantum Triple Ratchet, near-zero metadata, open source, non-profit, no ads
2	Nostr DMs (NIP-17)	6.6	No operator, no ads, sender hidden from relays and fully open code, but no forward secrecy, legacy NIP-04 clients still leak metadata, and a lost key is unrecoverable
3	Filen Chats	5.8	German zero-knowledge chat inside an encrypted cloud: E2EE by default, no ads, open clients and outside the CLOUD Act, but text-only, no per-message key rotation and no published audit
4	MEGA Chat	5.7	E2EE by default and no ad business, but older sender-key design without per-message forward secrecy, a 2022–23 history of cryptographic breaks, and routine subscriber-data disclosure
5	WhatsApp	5.0	Strong Signal-protocol content encryption undermined by Meta metadata harvesting, ads, repeated spyware targeting and unresolved insider-access allegations
6	Telegram Secret Chats	4.9	Opt-in E2EE (MTProto 2.0) that is mobile-only, single-device, no groups; the rest of the account still lives on Telegram's servers
7	Viber	4.7	Default E2EE for chats, groups and 1:1 calls since 2016, outside the CLOUD Act, but ad-funded with third-party trackers, closed protocol and phone numbers exposed in groups
8	Arattai	4.7	Mandatory E2EE for direct chats (Nov 2025) and groups (Sep 2026), ad-free and Zoho-run, but undisclosed protocol, no audit and Indian traceability rules
9	Facebook Messenger	4.4	Default E2EE for personal 1:1 chats (Signal protocol plus Labyrinth encrypted backups), but bound to the Facebook identity and social graph that feed Meta's ad business; Marketplace, business and Community Chats are not end-to-end encrypted
10	Google Messages (RCS)	4.4	Default Signal-protocol E2EE between Android users and MLS E2EE with iPhone since May 2026, but silent fallback to plain SMS and Google holds all metadata
11	Zoom Team Chat	3.8	Optional E2EE for chat (Advanced Chat Encryption) and post-quantum E2EE for meetings, but both are off by default and the employer controls the account
12	WhatsApp Business	3.6	E2EE to the business endpoint; with the Cloud API, Meta holds the keys and decrypts on the business's behalf
13	Telegram Cloud Chats	3.4	Server-side encryption only; Telegram can read every cloud message and has handed IP addresses and phone numbers to authorities on tens of thousands of occasions since late 2024
14	Zoho Cliq	3.2	Ad-free and region-hosted in Zoho-owned data centres, but no E2EE: messages are encrypted in transit and at rest and readable server-side for DLP and e-discovery
15	Botim	3.2	Claims E2EE for chats and calls with key verification, but no protocol or audit, cross-app tracking identifiers, fintech identity data, and a UAE licence granted by a regulator that blocks unlicensed encrypted VoIP
16	Discord	3.1	Voice and video E2EE by default via the audited DAVE protocol; all text readable by Discord; 2025 vendor breach exposed about 70,000 ID photos
17	Microsoft Teams Chat	3.1	Chat never E2EE; E2EE only for opt-in 1:1 calls and Premium meetings; built for employer audit and retention

Rank	Service	Weighted score (/10)	One-line verdict
18	Steam Chat	3.0	Text encrypted only in transit and readable by Valve; no ad network, but no published protocol or audit
19	Google Chat	2.8	No E2EE; client-side encryption exists for Meet, Drive and Gmail but not Chat; readable by Google and Workspace admins
20	Slack	2.7	No E2EE anywhere; Enterprise Key Management gives customers the master key but Slack still decrypts server-side; customer data trains non-generative global models unless an admin emails an opt-out
21	X Chat	2.5	Marketed as E2EE, but private keys sit on X servers behind a 4-digit PIN, no forward secrecy, no MITM protection, and the promised whitepaper never appeared
22	Reddit Chat	2.5	Server-readable DMs on a pseudonymous account; public content licensed for AI training, private messages excluded by policy
23	LinkedIn Messaging	2.0	No E2EE, real-name professional identity, expanded data sharing with Microsoft affiliates since November 2025
24	Instagram DMs	1.7	E2EE removed on 8 May 2026; every DM now readable by Meta and linked to its ad identity

Eight findings matter most. First, the gap between WhatsApp and Signal is not the cipher (both use the Signal protocol for content) but everything around it: who runs the server, what metadata it keeps, and what it does with it. Second, Telegram is two different products: the default cloud chat is a server-readable messenger, and the secret chat is a niche feature most users never enable. Third, MEGA Chat is a credible E2EE option that privacy comparisons usually ignore, but it inherits MEGA's account-key architecture, which ETH Zurich researchers broke in 2022 and again in 2023 before patches landed. Fourth, Facebook Messenger now encrypts personal chats by default with the same Signal protocol, yet scores below WhatsApp because it is bound to a real-name Facebook profile and Meta's advertising graph. Fifth, none of the social and gaming platforms encrypts text DMs end-to-end, and Instagram removed its opt-in encryption in May 2026, so all five belong in the bottom tier. Sixth, workplace and business tools add a second reader, the employer or the business: Teams, Slack and Zoho Cliq chat are never end-to-end encrypted, Zoom encrypts only when admins opt in, and WhatsApp Business via the Cloud API lets Meta hold the keys. Seventh, calls are better protected than text: nine services encrypt one-to-one calls end-to-end by default, while X Chat, which markets itself as encrypted, keeps users' private keys on its own servers. Eighth, Nostr shows what removing the operator buys and costs: no company to subpoena or to sell ads, but no forward secrecy and no recovery if the key is lost.

Methodology and scoring

Each configuration is scored 0–10 on eight criteria, then combined with the weights below to a single / 10 figure. Weights favour what protects message content and identity against the operator itself, since that is the threat model that separates these apps.

Criterion	Weight	What a 10 looks like	What a 0 looks like
Content encryption	20%	E2EE on by default for all chat types, modern ratcheting protocol, post-quantum protection, formally analysed	No E2EE; operator can read messages
Metadata minimisation	15%	Operator cannot see who talks to whom; contact discovery is private; nothing retained beyond account existence	Full social graph, timestamps, IPs, device data retained and profiled
Third-party data sharing and monetisation	10%	No ads, no data sharing with affiliates or brokers	Metadata feeds an advertising business across a corporate group

Criterion	Weight	What a 10 looks like	What a 0 looks like
Legal disclosure posture and jurisdiction	10%	Little to hand over, documented in transparency reports; no coercive home jurisdiction	Routine bulk disclosure of identifying data; opaque reporting
Openness and verifiability	15%	Client and server open source, reproducible builds, public protocol specs, independent audits and academic analysis	Closed source; protocol claims cannot be checked
Backups, multi-device and cloud exposure	10%	Any cloud copy is E2EE with keys the operator never sees; multi-device without weakening guarantees	Plaintext-equivalent copies of chats held by the operator or a third party
Track record (2024–2026)	10%	Vulnerabilities rare, disclosed and patched within days; no credible insider-access claims	Repeated exploitation, slow fixes, or unresolved allegations of operator access
Secure-by-default usability	10%	The private mode is the only mode; users cannot accidentally fall back to something weaker	Privacy features are opt-in, buried, or unavailable on some platforms

Telegram is scored twice because its two chat types have different security properties, and most users only ever use the default. The direct-message features of Discord, Steam, Reddit, LinkedIn and Instagram, and the workplace tools Teams, Slack, Zoho Cliq, Zoom and WhatsApp Business, and Viber, Arattai, Botim, X Chat, Google Chat, Nostr DMs, Google Messages and Filen Chats, are scored on the same eight criteria so every service sits on one scale. Sources are limited to primary documents (vendor security pages, transparency reports, court filings, peer-reviewed papers) and established security press; every figure in this report links to the page it was taken from, and claims that remain allegations are labelled as such.

Encryption architecture

Only Signal protects both the first handshake and every later message against a future quantum computer; Telegram cloud chats are the only configuration where the operator holds the keys.

Property	WhatsApp	Facebook Messenger	Telegram Cloud Chat	Telegram Secret Chat	Signal	MEGA Chat
E2EE by default	Yes, all 1:1, groups, calls	Yes for personal 1:1 chats and calls since the Dec 2023 rollout; not for Marketplace, business or Community Chats (Meta list)	No — client-server encryption only; messages decrypted on Telegram servers	Yes, but opt-in per chat	Yes, all chats and calls	Yes, all chats and calls
Protocol	Signal Protocol (X3DH + Double Ratchet), WhatsApp's closed implementation	Signal Protocol (Meta implementation) for messages; Labyrinth for encrypted message storage (whitepaper)	MTProto 2.0 client-server (AES-256 IGE)	MTProto 2.0 secret chats: DH key exchange, AES-256 IGE, SHA-256	Signal Protocol with PQXDH handshake and Triple Ratchet (Double Ratchet + SPQR)	Strongvelope: sender keys wrapped with x25519 ECDH, Ed25519 signatures, AES-128 content encryption

Property	WhatsApp	Facebook Messenger	Telegram Cloud Chat	Telegram Secret Chat	Signal	MEGA Chat
Forward secrecy	Per-message (Double Ratchet)	Per-message (Double Ratchet) in E2EE chats	None against operator (operator holds plaintext)	Periodic re-keying after 100 messages or one week	Per-message, classical and post-quantum	No per-message ratchet; keys change only on membership change or manual key rotation (design doc lists further guarantees as not supported)
Post-compromise security	Yes (DH ratchet)	Yes in E2EE chats (DH ratchet)	No	Partial (re-key)	Yes, classical and post-quantum	Limited to key rotation events
Post-quantum protection	None announced for messaging	None announced	None	None	PQXDH since 2023; SPQR rolling out since Oct 2025 with downgrade removal planned once clients converge	None
Group chats	E2EE, Sender Keys	Personal group chats may be E2EE (check per chat); Facebook Community Chats are not	Server-side only	Not available — secret chats are 1:1 only	E2EE, Sender Keys with private group state	E2EE, sender keys; optional key rotation to make a group closed
Multi-device	Up to 4 linked devices, E2EE per device	Mobile and web (Chrome/Edge) E2EE per device; desktop apps retired Dec 2025	All devices, via server	Single device only; chats do not sync	Linked devices, E2EE per device	All devices, via encrypted cloud store
Independent analysis	Protocol is academically analysed; WhatsApp's binary is not open	Meta-published whitepapers only; closed source; key transparency for 1:1 chats since Nov 2025	MTProto 2.0 analysed by Royal Holloway (2021–22), several fixable flaws found	Same MTProto analyses; secret-chat layer rarely studied	Most analysed messaging protocol; SPQR modelled in ProVerif; two integrity attacks published March 2026, patched in 2 and 8 days	Design docs published 2015; account crypto broken by ETH Zurich 2022 and Eurocrypt 2023, including recovery of the Curve25519 chat key; patched

The Signal protocol's quantum step is the material 2025–26 change: Signal's September 2025 blog post describes SPQR as a sparse post-quantum ratchet layered on the Double Ratchet to form a Triple Ratchet with forward secrecy and post-compromise security that hold even if elliptic-curve cryptography is broken, with keys packed into 64-byte chunks so data use stays flat, and with a planned later update that forces SPQR on all sessions. WhatsApp runs the same classical Signal protocol but had announced no post-quantum work for messaging as of this report.

MEGA's Strongvelope protocol is older in design: one sender key per participant is encrypted to each recipient with a static x25519 chat key, so compromising a user's chat key exposes their past messages in a way the ratcheting protocols do not. MEGA's own key authentication paper confirms the Ed25519 identity key, x25519 chat key and RSA sharing key hierarchy; the ETH Zurich attacks showed that this hierarchy fell when the master key was recoverable.

Metadata, data sharing and government disclosure

Signal can produce only an account's creation date and last-connection date under subpoena; Telegram hands over phone numbers and IP addresses at scale; WhatsApp feeds account and device metadata into Meta's advertising group.

Dimension	WhatsApp	Facebook Messenger	Telegram (both chat types)	Signal	MEGA Chat
Identifier required	Phone number (usernames rolling out 2026)	Facebook account (real-name policy), plus phone or email	Phone number	Phone number at sign-up; usernames hide it from contacts	Email address
Contact discovery	Uploads address-book hashes to Meta servers	Facebook friend graph, plus optional phone-contact upload	Uploads contacts to Telegram servers (optional)	Private contact discovery in SGX enclaves; server never learns the list	Manual invites by email or link
Who-talks-to-whom visible to operator	Yes (routing metadata, group membership)	Yes, linked to the full Facebook profile and social graph	Yes, plus message content for cloud chats	Hidden by Sealed Sender; group membership encrypted	Yes (chat participants, membership, timestamps)
Shared with corporate affiliates	Yes — account, device and Updates-tab activity shared with Meta companies; ads in Updates tab since 2025, EU rollout paused pending the Irish DPC	Yes — Messenger activity falls under Meta's general privacy policy, which uses account and activity data for ads across Meta products	Telegram Ads in channels; no corporate group	None — Signal Foundation is a non-profit with no ad business	None disclosed; MEGA is subscription-funded
Transparency reporting	Meta transparency centre (aggregate, Meta-wide)	Meta transparency centre (aggregate, Meta-wide)	Quarterly bot-based reports, per country, viewable only from an account in that region	Published government requests with court documents	Semi-annual report; 15th edition issued 30 March 2026
What is handed over on a valid order	Subscriber info, IP logs, contacts and routing metadata; message content unavailable	Subscriber info, IP logs, metadata, and content of chats outside E2EE (Marketplace, business, Community, pre-upgrade history)	Phone number and IP address when a judicial order names a suspect of a crime (policy since Sept 2024)	Account creation date and last connection timestamp only	Subscriber details and IP data; chat content unavailable
Home jurisdiction	USA (Meta Platforms); Irish entity for EU	USA (Meta Platforms); Meta Platforms Ireland for EU	Dubai operations; founder indicted in France (Aug 2024) and under judicial supervision	USA (non-profit)	New Zealand historically; the report of March 2026 states the contractual provider became MEGA Privacy LLC, Hungary, in March 2025

The Telegram numbers are the largest. Telegram's own regional transparency data, as reported by TechCrunch in January 2025, showed phone numbers and IP addresses given to US authorities on 900 occasions in 2024 affecting 2,253 users, to Indian authorities on 14,641 occasions affecting 23,535 users, and to UK authorities 142 times affecting 293 users. For the first quarter of 2025 alone, Telegram reported data on 22,777 users handed over, against 5,826 in the same quarter of 2024 (Mein-MMO

summary of the Q1 2025 report). Before the September 2024 policy change, Telegram disclosed data only in terrorism cases.

WhatsApp's exposure is structural rather than episodic. TechCrunch notes that since 2016 the privacy policy has permitted sharing metadata such as phone numbers and device information with Facebook, and the 2025 ad launch uses account information, device metadata, approximate location and Updates-tab activity for targeting (TechSpot). Meta states phone numbers are removed or altered before data reaches Meta for ads. Content remains unreadable to Meta under the published design; the dispute in US courts (next sections) is about whether that design is honoured internally.

MEGA's disclosure volume is material for a privacy-marketed service: its 2020 report, as summarised by TechNadu, counted roughly 9,100 subscriber-information requests in twelve months with only seven declined, and MEGA retains account data for 12 months after closure. The current report states that details are routinely shared with relevant law-enforcement authorities. Chat content is never among the items disclosed, because MEGA does not hold the keys.

Server locations and legal jurisdictions

Where the servers sit matters less than which law binds the operator: WhatsApp and Signal are both reachable by US process under the CLOUD Act regardless of data-centre country, Telegram's data sits in rented space it controls under BVI and Dubai entities, and MEGA now answers to a Hungarian controller with no US-hosted data.

Factor	WhatsApp	Facebook Messenger	Telegram	Signal	MEGA Chat
Operating legal entities	WhatsApp LLC (Delaware, USA) for most of the world; WhatsApp Ireland Limited for EEA/UK users; both owned by Meta Platforms, Inc.	Meta Platforms, Inc. (USA) for most of the world; Meta Platforms Ireland Limited for EEA/UK users	Telegram Messenger Inc. (contracting entity); parent Telegram Group Inc. and Telegraph Inc. (British Virgin Islands); Telegram FZ-LLC (Dubai, UAE) per the privacy policy	Signal Foundation, a California 501(c)(3), and Signal Messenger LLC, a California LLC, both in Mountain View (Sota analysis)	MEGA Privacy LLC (Hungary) is the controller for MEGA Cloud and S4 users; Mega Privacy (NZ) Limited covers standalone VPN and Pass plans; Mega Limited (Auckland) remains the group company (privacy policy, April 2026 edition)
Primary server locations	Meta-owned data centres, mainly United States, with European sites (Ireland, Sweden, Denmark) and Singapore; location is Meta's choice, not the user's	Same Meta-owned data centre fleet as WhatsApp (mainly United States, plus European and Asian sites)	Telegram-owned servers in rented third-party data centres across several jurisdictions; EEA/UK sign-ups stored in the Netherlands; encryption keys for stored data held in other data centres in different jurisdictions	Rented cloud: AWS in North America for most hosting and message queuing (DynamoDB), plus Google Cloud and Cloudflare; Azure confidential computing used for enclave services (Taiwan MODA interview)	MEGA-owned servers in primary data centres in Japan, Canada and the European Union (New Zealand dropped from the 2026 policy); policy states no personal data is stored in or made available from the United States (server list)

Factor	WhatsApp	Facebook Messenger	Telegram	Signal	MEGA Chat
Governing law for users	US law; Irish law and GDPR for EEA/UK users	US law; Irish law and GDPR for EEA/UK users	Terms governed from the BVI/Dubai structure; GDPR applies to EEA/UK data via standard contractual clauses, since neither the BVI nor the UAE has an EU adequacy decision (ConductAtlas)	US (California) law	Hungarian/EU law and GDPR for Cloud users; New Zealand Privacy Act 2020 for NZ-entity services
US CLOUD Act reach	Yes — Meta is a US company, so US warrants reach data wherever stored, including Irish servers	Yes — Meta is a US company; reach extends to all non-E2EE content and metadata	No direct reach; cooperation runs through judicial requests Telegram chooses to honour, and via the EU DSA channel for EU authorities	Yes in law, but Signal holds only account creation and last-connection timestamps, so compelled production yields almost nothing	No direct reach: no US entity, no US-hosted data; US requests go via NZ or Hungarian mutual-assistance process
EU regulator	Irish Data Protection Commission (lead supervisory authority); EU DSA very-large-platform obligations	Irish Data Protection Commission; EU DSA very-large-platform obligations	EU DSA applies; Telegram has no EU establishment, so enforcement runs through its EU legal representative	No EU establishment; GDPR applies to EEA users extraterritorially	Hungarian National Authority for Data Protection (NAIH) as lead authority since the 2025 restructuring
Intelligence-alliance exposure	Five Eyes (USA)	Five Eyes (USA)	None of the hosting entities is in a Five Eyes state; UAE and BVI have their own disclosure regimes and limited judicial transparency	Five Eyes (USA), mitigated by data minimisation	Hungary is EU but not Five Eyes; New Zealand (group parent) is Five Eyes
What the jurisdiction can actually obtain	Metadata, contacts, IP logs, unencrypted backups via Apple/Google; no content	Facebook profile, social graph, metadata, IP logs, and readable content of non-E2EE chat types; no content from E2EE personal chats	Full cloud-chat content, contacts, IP and phone number; secret-chat content unavailable	Two timestamps; nothing else exists	Account data, IP logs, usage metadata; no chat or file content

Two points deserve emphasis. First, server geography is a weak proxy for safety: a US company's European data centre is still reachable by a US warrant (Fast LTA explainer), which is why WhatsApp's Irish entity does not change its exposure and why Signal's protection comes from holding nothing rather than from where it hosts. Second, Telegram's structure, a BVI parent, a Dubai operating company, Netherlands servers for Europeans and keys split across unnamed jurisdictions, is designed to make any single state's process slow, which cut both ways in 2024: it frustrated French investigators until Durov's arrest, after which the company began honouring requests at scale.

For MEGA the 2025–26 change is the one to watch. Moving the data controller from New Zealand to a Hungarian LLC places European users under GDPR with a Hungarian lead regulator, keeps data out of the US, and removes New Zealand from the primary storage list. Hungary's data-protection authority has a mixed independence record within the EU, and the rationale for the move has not been published, so users who chose MEGA for its New Zealand domicile should re-evaluate on their own threat model.

Backups, multi-device and cloud exposure

A chat is only as private as its weakest copy: WhatsApp's default cloud backup still sits unencrypted in Google Drive or iCloud unless the user turns on the encrypted option, while Signal's new backups are encrypted with a key the server never holds.

Aspect	WhatsApp	Facebook Messenger	Telegram Cloud Chat	Telegram Secret Chat	Signal	MEGA Chat
Default backup state	Google Drive / iCloud backup, not E2EE unless user enables end-to-end encrypted backups	Secure Storage: E2EE server-side history via Labyrinth, protected by PIN, 40-character code, or Apple/Google account	Entire history lives on Telegram servers by design	No backup; chat exists only on the one device	No cloud copy unless user opts in	History stored on MEGA servers, encrypted client-side under the user's keys
Encrypted backup option	Yes, opt-in, 64-digit key or password	Built in; Labyrinth 1.1 (May 2026) writes messages to the encrypted backup as they are sent	Not applicable	Not applicable	Yes — Secure Backups, opt-in, 64-character recovery key generated on device, free 45-day text tier, paid media tier; all platforms from Signal 8.0, Feb 2026	Built in; no separate option
Who can read the backup	Google/Apple and anyone with a lawful-access route to them, unless E2EE backup is on	Nobody without the recovery method; if recovery uses an Apple or Google account, that account becomes part of the trust boundary	Telegram	Nobody else	Nobody; backup is unlinked to account or payment	MEGA cannot; recovery key or password required
Account recovery risk	SIM-swap plus restored backup yields full history; two-step PIN mitigates	Facebook account takeover adds an attacker device; resetting Secure Storage permanently deletes history	SIM-swap yields full account unless 2FA password set	Device loss destroys chat	Registration Lock PIN; losing recovery key loses backup	Losing password and recovery key loses everything, including chats
Chat partner's copy	Other party's backup may be unencrypted	Partner's history is also held in E2EE Secure Storage	Server-side regardless	Both devices only	Other party's backup is E2EE if they back up at all	Encrypted under their keys

Two details reshape the comparison. Signal's backup design has a known trade-off, raised in community review when SPQR launched: a backup is encrypted under a single static key, so forward secrecy does not extend to the backup set (Algustionesa analysis). Signal accepts this because the alternative is no recovery at all, and the key never touches its servers.

For WhatsApp, the practical leak is the counterpart's device: a user who enables encrypted backups and disappearing messages is still exposed if the person they message keeps a plain iCloud backup. Telegram secret chats avoid this entirely, at the price of no desktop access and no history on a new phone. MEGA sits in between: everything is recoverable from any device, but only with the account password, which is also the root of the account's key hierarchy, so password strength is the entire security boundary.

Openness, verifiability and ownership

Signal is the only service where both client and server code are public and builds are reproducible; WhatsApp is the only one where the user has no way to verify any encryption claim beyond trusting Meta.

Factor	WhatsApp	Facebook Messenger	Telegram	Signal	MEGA Chat
Client source	Closed	Closed	Open (GPL), reproducible builds for Android and iOS	Open (AGPL), reproducible Android builds	Public source for web, desktop, mobile and MEGAchat SDK
Server source	Closed	Closed	Closed	Open (AGPL)	Closed
Protocol specification	Security whitepaper	E2EE overview and Labyrinth whitepapers	MTProto docs	Full specifications with formal analyses	2015 design docs (Strongvelope, key authentication) and a security whitepaper
Key verification	Safety codes; key transparency directory since 2023	Automatic key transparency for 1:1 chats (Nov 2025) plus manual key checks	Key visualisation in secret chats only	Safety numbers; usernames	Fingerprint comparison of Ed25519 identity key
Independent audit or academic review	Protocol yes; implementation no public audit	No public independent audit of the implementation	Royal Holloway MTProto analysis; no public third-party audit of secret chats	Continuous academic scrutiny; Cure53 and NCC audits of components; March 2026 IACR attacks disclosed and fixed	ETH Zurich (2022) and Albrecht et al. (2023) attacks; no published independent audit of MEGAchat since
Ownership and funding	Meta Platforms, advertising revenue	Meta Platforms, advertising revenue	Private company controlled by Pavel Durov; Premium subscriptions, ads, TON ecosystem	Signal Foundation (501(c)(3)) and Signal Messenger LLC; donations	MEGA Limited (NZ) with Hungarian provider entity since March 2025; paid storage plans
Encryption policy stance	Publicly defends E2EE; litigated against NSO and won $167M (2025)	Told the US Senate in 2024 it would not offer an opt-out from default E2EE; link safety uses on-device and blinded server checks	Founder indicted in France Aug 2024 partly for non-cooperation; policy loosened a month later	Has stated it will exit markets rather than weaken encryption	Declares zero tolerance for illegal content while defending user-controlled keys

Openness is where Telegram's reputation and reality diverge most. Its clients are open and reproducible, which is genuinely better than WhatsApp, but the server that stores and decrypts every cloud chat is closed, so the openness buys nothing for the default chat type. InCyber's analysis quotes cryptographer Matthew Green's complaint that Telegram has left its end-to-end encryption user experience essentially unchanged since 2016 while marketing itself as a secure messenger.

For MEGA, the public source is real and auditable, which is exactly how the ETH Zurich team found the 2022 flaws by reading it. That cuts both ways: the problems were found and fixed, but the Eurocrypt 2023 follow-up showed MEGA's first round of sanity-check patches could themselves be exploited to recover a user's RSA private key, suggesting the architecture rather than any single bug was the weakness.

Track record 2024–2026

WhatsApp has the most incidents of the five services, including two zero-click spyware campaigns and a still-live legal fight over insider access; Signal's two 2026 protocol bugs were fixed within days and its other headline exposure was an Apple OS bug.

Date	Service	Incident	Status
Jul 2026	WhatsApp	US District Judge Rita Lin dismissed the January class action for insufficient whistleblower detail, denied Meta's sanctions motion, and allowed refiling by August; the US Commerce Department had closed its own probe three months earlier	Amended complaint expected; Texas AG suit filed May 2026 ongoing
May 2026	Facebook Messenger	Labyrinth 1.1 rolled out so messages reach the encrypted backup as sent, fixing history loss on device change	Shipped
Apr 2026	Signal	FBI recovered deleted Signal messages from a defendant's iPhone via iOS notification storage; Apple patched CVE-2026-28950 on 22 April 2026	Fixed in iOS 26.4.2 / 18.7.8; flaw was Apple's, not Signal's
Mar 2026	Signal	IACR paper disclosed two integrity attacks: a malicious server could inject messages via the username-resolution flow and via Sealed Sender (latent since 2018)	Patched 2 and 8 days after disclosure (Android fix 28 Jan 2026)
Mar 2026	Facebook Messenger	Advanced Browsing Protection checks links in E2EE chats against server threat data without revealing full URLs	Shipped
Jan 2026	WhatsApp	Class action filed by users in six countries alleging Meta staff and contractors can read messages via an internal tool; Meta called it categorically false	See Jul 2026 row
Jan 2026	WhatsApp	Strict Account Settings lockdown mode launched days after the lawsuit	Shipped
Nov 2025	Facebook Messenger	Key transparency added for automatic key verification in 1:1 chats	Shipped
Sep 2025	WhatsApp	Former head of security Attaullah Baig sued Meta alleging about 1,500 engineers had unrestricted access to user metadata and that 100,000 daily account takeovers went unaddressed (Analytics Insight)	Litigation ongoing; allegations unproven
Aug 2025	WhatsApp	CVE-2025-55177 linked-device sync flaw chained with Apple CVE-2025-43300 in a zero-click spyware campaign against roughly 200 users	Patched; added to CISA KEV
Jan–Mar 2025	WhatsApp	Paragon Graphite zero-click exploit via malicious PDF; about 90 journalists and civil-society targets notified 31 Jan 2025 (Cybersecurity News)	Mitigated server-side; Paragon later cut off Italy
Jan 2025	Facebook Messenger	Meta said default E2EE had reached the vast majority of users, 13 months after the Dec 2023 announcement (iTechGuides timeline)	Rollout effectively complete; exceptions remain
Jan 2025	Telegram	Transparency data showed sharp rise in fulfilled data requests after the September 2024 policy change	Ongoing; Q1 2025 alone covered 22,777 users
Sep 2024	Telegram	Privacy policy changed to allow disclosure of IP and phone number for any criminal investigation with a judicial order, not just terrorism (AlternativeTo)	In force
Aug 2024	Telegram	Pavel Durov arrested at Le Bourget and indicted in France, including for failure to comply with law-enforcement requests; placed under judicial supervision	Case ongoing
Mar 2023	MEGA	Caveat Implementor: new RSA key-recovery attacks exploiting MEGA's post-2022 sanity checks	Patched
Jun 2022	MEGA	Five attacks by ETH Zurich allowing a malicious server to recover node keys, the Ed25519 signing key and the Curve25519 chat key; MEGA patched three immediately	Patched; one legacy chat-key oracle deemed impractical to exploit

The two MEGA rows pre-date the 2024–2026 window but are included because no comparable public analysis has been performed since, so they remain the most recent independent assessment of MEGA's cryptography. The WhatsApp insider-access allegations have not been proven in any court; what the record does establish is that a federal judge declined to call them frivolous and that a separate state attorney general has filed similar claims.

Social and gaming platform messaging

None of these five platforms end-to-end encrypts text messages in October 2026, so the operator can read every DM; Discord is the only one with default E2EE anywhere, and only for voice and video. They score 1.7 to 3.1 out of 10 on the same eight criteria used above.

Property	Discord	Steam Chat	Reddit Chat	LinkedIn Messaging	Instagram DMs
Operator	Discord Inc. (USA, private)	Valve Corporation (USA, private)	Reddit, Inc. (USA, listed)	LinkedIn Corporation, a Microsoft subsidiary (USA); LinkedIn Ireland for EU	Meta Platforms (USA); Meta Platforms Ireland for EU
Text messages E2EE	No — Discord says text is not E2EE and it has no plans to change that	No — Valve staff: encrypted to and from servers only	No	No	No — opt-in E2EE removed 8 May 2026
Voice and video E2EE	Yes, by default — DAVE protocol (MLS-based) enforced on all calls since March 2026, except Stage channels	Voice described as E2EE and peer-to-peer since the 2018 chat rebuild; no published protocol	Not applicable	Not applicable	No
Who can read DMs	Discord (moderation, Trust & Safety, legal process)	Valve	Reddit	LinkedIn / Microsoft	Meta, including previously encrypted chats migrated to Meta-readable storage
Identity required	Email; pseudonymous username; phone optional	Email; pseudonymous; tied to purchase and payment history	Email optional; pseudonymous	Real name, employer, work history	Account tied to Instagram profile, contacts and Meta ad identity
Ads and data use	Ads and sponsored Quests; no corporate parent sharing	Store business; no ad network	Ads; public posts licensed to Google and OpenAI for AI training, CEO said private messages are excluded (Bloomberg via BNN)	Data shared with Microsoft affiliates for ads from 3 Nov 2025 outside EU/UK/CH; profile and posts used for AI training, private messages excluded	Meta ads across Facebook, Instagram, Messenger, WhatsApp; DM content now technically available to Meta
Openness	DAVE protocol open source and audited by Trail of Bits; clients and servers closed	Closed	Closed	Closed	Closed
Jurisdiction and CLOUD Act	USA; CLOUD Act applies	USA; CLOUD Act applies	USA; CLOUD Act applies	USA (Microsoft); CLOUD Act applies; Irish DPC for EU	USA (Meta); CLOUD Act applies; Irish DPC for EU

The incident record for these platforms in the 2024–2026 window is short but telling.

Date	Service	Incident	Status
May 2026	Instagram DMs	Meta removed end-to-end encryption from Instagram DMs, citing low opt-in use and pointing users to WhatsApp; the Global Encryption Coalition urged a reversal	In force since 8 May 2026
May 2026	Discord	DAVE E2EE confirmed as default for every voice and video call; unencrypted fallback being removed	Shipped; text excluded
Mar 2026	Instagram DMs	Meta announced the E2EE shutdown weeks after a New Mexico jury verdict against Meta on platform safety	See May 2026 row
Nov 2025	LinkedIn	Data sharing with Microsoft affiliates for ads and EU/EEA/CH data for AI training began 3 November (Gulf News); private messages excluded	In force; opt-out available
Sep–Oct 2025	Discord	Breach of a third-party support vendor exposed names, emails, support messages and about 70,000 government ID photos submitted for age appeals; attackers claimed far more	Vendor access revoked
Jan 2025	LinkedIn	Class action alleging Premium InMail messages were used for AI training was dropped nine days after filing after LinkedIn showed evidence they were not	Dismissed without prejudice

Discord

Discord is the strongest of the five for calls and among the weakest for text. Its DAVE protocol is open, externally audited and now mandatory, so voice channels, group calls and Go Live streams are end-to-end encrypted by default. Text in DMs, group DMs and servers is not, and Discord has said it has no plans to change that because moderation, search and bots depend on server-side access. The October 2025 vendor breach is the sharper warning: age-verification ID images held by a support contractor were stolen, which shows that the riskiest data on a chat platform is often not the chat.

Steam Chat

Steam Chat is a convenience feature on a store, not a private messenger. Valve has stated that text is encrypted only between client and server, so Valve holds readable history, and no protocol or audit has been published for the voice path it describes as end-to-end encrypted. On the positive side, Valve runs no advertising network and no identity is required beyond an email, but chats sit alongside a detailed record of purchases, playtime and payment data.

Reddit Chat

Reddit Chat and legacy private messages are server-readable and tied to a pseudonymous account. Reddit licenses public content to Google and OpenAI for AI training; its CEO has said private messages, emails and phone numbers are excluded from those deals, but that is a policy promise rather than a technical one. Pseudonymity is its main privacy advantage over LinkedIn and Instagram.

LinkedIn Messaging

LinkedIn messages are readable by LinkedIn and linked to a real name, employer and professional history, which makes them the most identity-revealing DMs in this report after Instagram. Since November 2025 LinkedIn shares more member data with Microsoft affiliates for ads in most regions and uses profile and post data for AI training in Europe, with private messages excluded by policy. The January 2025 lawsuit claiming InMail was used for AI training was withdrawn after LinkedIn produced evidence to the contrary. Treat LinkedIn messages as business correspondence your employer, a recruiter or a court could see.

Instagram DMs

Instagram is the only service in this report that has moved backwards. Opt-in end-to-end encryption, available since December 2023, was removed on 8 May 2026, and previously encrypted chats were migrated to storage Meta can read. Meta cited low adoption; critics, including the Global Encryption Coalition, argued that low adoption of an opt-in feature is a case for making it default. Every Instagram DM is now available to Meta for moderation, legal process and, potentially, ads or AI, so anything sensitive belongs on WhatsApp or Signal instead.

Other consumer messengers: Viber, Arattai, Botim, X Chat and Google Chat

Viber and Arattai are the only two of these five with default end-to-end encryption for text; Botim claims it but is licensed in a jurisdiction that blocks unlicensed encrypted calling; X Chat stores users' private keys on X's own servers; Google Chat has no end-to-end encryption at all.

Property	Viber	Arattai	Botim	X Chat	Google Chat
Operator	Viber Media S.à r.l. (Luxembourg), owned by Rakuten (Japan)	Zoho Corporation (India, privately held)	Astra Tech (UAE); TDRA-licensed VoIP app	X Corp. (USA)	Google LLC (USA); Google Ireland for EU
Text messages E2EE	Yes by default for 1:1 and group chats since 2016 (Viber); own protocol, closed source	Yes — mandatory for direct chats since 18 Nov 2025 and for groups since 25 Sep 2026; protocol undisclosed	Claimed — Botim FAQ says messages and calls are E2EE with per-chat keys and a Verify Encryption Key screen; no protocol documentation or audit	Claimed — libsodium sealed boxes, but private keys are stored on X's servers under a 4-digit PIN via the Juicebox protocol, with no man-in-the-middle protection and no forward secrecy (Matthew Green); old DMs stay unencrypted	No — TLS and encryption at rest; Google's client-side encryption covers Drive, Docs, Gmail, Calendar and Meet, not Chat messages
Who can read messages	Nobody but participants; Viber servers see name, number, device and activity	Participants only, if Zoho's implementation holds; no independent review published	Participants only if the claim holds; TDRA licensing exists because UAE regulators block unlicensed encrypted VoIP	X, in effect: whoever controls the Juicebox key store and the PIN-derivation path; X's own help page says a malicious insider or legal process could compromise a conversation undetected	Google, Workspace admins and compliance tools
Identity required	Phone number; number visible to all members of a group	Phone number	Phone number; wallet and payments add KYC identity	X account (often real identity); Premium for larger files	Google account; Workspace identity at work
Ads and data use	Ad-supported, roughly a fifth of revenue; third-party ad networks and attribution SDKs in the app (Mozilla PNI)	Ad-free (Zoho)	Apple privacy label lists identifiers used to track across apps; super-app with payments, remittances and shopping	X's business is ads; X says Chat itself has no ads or tracking	Consumer: Google account data; Workspace: no ads on work content
Openness	Closed; encryption overview published	Closed; no protocol documentation	Closed; none	Closed; whitepaper and open source promised in 2025, not delivered by April 2026 (Tesorb)	Closed

Property	Viber	Arattai	Botim	X Chat	Google Chat
Jurisdiction and CLOUD Act	Luxembourg / EU; Japanese parent; outside CLOUD Act	India; Indian traceability rules for large messaging platforms	UAE telecom law; TDRA oversight; licensed because it meets regulator requirements	USA; CLOUD Act	USA; CLOUD Act; Irish DPC for EU

Viber

Viber has had default end-to-end encryption for one-to-one and group chats and one-to-one calls since 2016, with password-protected hidden chats and disappearing messages on top. Its weaknesses are the business model and the closed protocol: about a fifth of Rakuten Viber's revenue comes from ads served through third-party networks, phone numbers are exposed to every member of a group, group calls are only encrypted in transit, and the cryptography has not been independently audited in public. Its Luxembourg domicile and Japanese parent keep it outside the US CLOUD Act.

Arattai

Zoho's consumer messenger moved fast after its October 2025 surge in India: end-to-end encryption became mandatory for direct chats in November 2025, with old unencrypted threads archived, and arrived for group chats in September 2026. Calls were encrypted earlier. The gaps are maturity and transparency: the protocol is undisclosed, no independent audit has been published, and Indian law imposes traceability duties on large messaging platforms that sit uneasily with E2EE. Ad-free and Zoho-hosted, it is a credible option whose cryptography remains unverified.

Botim

Botim is the TDRA-licensed calling and messaging app that works where other VoIP services are blocked in the UAE, and it has grown into a super-app with payments, remittances and shopping. Its FAQ describes end-to-end encryption for messages and calls with per-chat keys and manual key verification. Two facts temper that claim: no protocol documentation or audit has been published, and the licence exists because the app satisfies a regulator whose stated policy is to block unlicensed encrypted VoIP. The fintech features add identity and transaction data that a pure messenger would not hold.

X Chat

X Chat, launched as a standalone app in April 2026, markets itself as end-to-end encrypted with "Bitcoin-style" encryption. Cryptographers reviewing it in 2025 found that user private keys are stored on X-run servers behind a four-digit PIN using the abandoned Juicebox protocol, that there is no forward secrecy, and that X's own help page admits a malicious insider or legal order could compromise a conversation without either party knowing. The promised whitepaper and open-source release had not appeared a year later. Treat X Chat as encrypted in transit with X able to recover keys, not as E2EE in the Signal sense.

Google Chat

Google Chat, free or in Workspace, encrypts in transit and at rest and nothing more. Google's client-side encryption programme, which puts keys in the customer's hands, covers Drive, Docs, Gmail, Calendar and Meet but has not been extended to Chat messages, so Google and Workspace administrators can read them. Calls go through Google Meet, where client-side encryption is available to Enterprise Plus customers and optional end-to-end encryption exists in the consumer Meet app. For work it belongs with Teams and Slack; for personal use it is the least private of the Google-branded messaging options.

Decentralised and carrier messaging: Nostr DMs and Google Messages (RCS)

Nostr has no operator to subpoena and hides the sender from relays, but its DMs lack forward secrecy and depend on which client you use; Google Messages encrypts RCS end-to-end with the Signal protocol and, since May 2026, across to iPhone with MLS, but silently falls back to unencrypted SMS and runs through Google's servers.

Property	Nostr direct messages	Google Messages (RCS)
Operator	None — open protocol; messages pass through independently run relays; clients such as Damus, Amethyst, Primal and 0xchat	Google LLC (USA); RCS backend is Google's Jibe for most carriers; Apple Messages on the other end since 2026
Text messages E2EE	Yes with NIP-17: NIP-44 encryption (secp256k1 ECDH, ChaCha20) wrapped in NIP-59 gift wraps sent from a disposable key; the older NIP-04 (deprecated) still exists in legacy clients and exposes sender and recipient	Yes by default between Google Messages users since 2021 (groups since 2023), using the Signal protocol; cross-platform E2EE over RCS with MLS rolling out with Apple since 11 May 2026; falls back to plain SMS/MMS when RCS is unavailable
Forward secrecy	No — static keys; a leaked private key (nsec) decrypts all past DMs; the MLS-based Marmot effort aims to fix this	Yes (Signal Double Ratchet; MLS for cross-platform)
Who can read messages	Participants only; relays see the recipient's public key, the gift-wrap's throwaway key and the sender's IP	Participants only for RCS chats marked encrypted; Google, carriers and anyone on the SMS path for fallback messages
Identity required	A key pair; no phone, email or account; optional NIP-05 name	Phone number and Google account
Metadata	Sender hidden from relays; recipient and timing (randomised) visible; no central social graph, but public keys are permanent identifiers	Google sees who messages whom, when, from which device; carriers see RCS routing
Ads and data use	None	Google ad business; Google states Messages content is not used for ads
Openness	Fully open protocol and open-source clients; NIP-44 audited by Cure53 (2023)	Signal protocol and MLS are open standards; Google Messages client is closed; technical paper published
Backups and multi-device	Encrypted events persist on relays; any device with the key can read; losing the key loses everything, no recovery	Android backups to Google are encrypted with the screen-lock PIN; SMS/MMS backups are not E2EE
Jurisdiction and CLOUD Act	No company; individual relays fall under their own countries' law and can be ordered to hand over encrypted events and IPs	USA; CLOUD Act applies to Google-held data

Nostr direct messages

Nostr is the only option in this report without a company behind it. With a modern client using NIP-17, messages are encrypted with NIP-44 and wrapped so that relays cannot see who sent them, and there is no operator to receive a subpoena, show ads or change the terms. The weaknesses are structural: there is no forward secrecy, so a single leaked private key exposes every past conversation; the deprecated NIP-04 scheme, which leaks sender and recipient, is still supported by many clients; relays see your IP unless you use Tor or a VPN; and your public key is a permanent identifier. Privacy therefore depends on client choice and key hygiene more than on the protocol's ceiling.

Google Messages (RCS)

Google Messages gives Android users default end-to-end encryption with the Signal protocol, and the GSMA's MLS-based standard, rolling out with Apple since May 2026, extends that to iPhone

conversations. The caveats are the carrier heritage: when a contact's phone is offline, roaming without data, or on an unsupported carrier backend, the conversation drops to SMS or MMS with no encryption, and Google's servers see the full metadata of who messages whom. It is a large upgrade over SMS and a reasonable default for everyday texting, not a replacement for Signal.

Encrypted-cloud chat: Filen Chats

Filen Chats is a German, zero-knowledge chat built into an encrypted cloud drive: end-to-end encrypted between Filen users by default, ad-free and outside the Five Eyes, but text-only, unaudited, and without the per-message key rotation that modern messengers use.

Property	Filen Chats
Operator	Filen Cloud Dienste UG (Recklinghausen, Germany); servers in German data centres (Filen)
Text messages E2EE	Yes by default — client-side encrypted between Filen users; AES-256-GCM content keys exchanged with per-account RSA-OAEP keys, per the open client code (Priviy review)
Forward secrecy and post-quantum	No per-message ratchet; keys tied to the account key pair; no post-quantum work
Calls	None — text, files and encrypted notes only
Who can read messages	Participants only, if the implementation holds; Filen servers see participants and timestamps
Identity required	Email address; no phone number
Ads and data use	None; paid storage plans fund the service
Openness	Clients open source (MIT) since 2023 on GitHub; server closed; the company said in November 2023 an independent audit was planned, and none has been published
Backups and multi-device	History stored encrypted on Filen servers under the account keys; losing the password and recovery key loses everything
Jurisdiction and CLOUD Act	Germany, GDPR and BDSG; no US entity, so outside CLOUD Act reach

Filen Chats

Filen Chats scores well for the same reasons MEGA Chat does and shares the same caveats. The German operator runs no ads, holds no keys and sits outside the CLOUD Act, and the open-source clients let anyone inspect the encryption. But the chat is a feature of a storage product rather than a messenger: it works only between Filen users, has no voice or video, and wraps message keys with each account's long-lived RSA key instead of ratcheting them, so a compromised account key exposes history. No independent audit has been published despite being promised in 2023. It is a sound choice for small teams already sharing files on Filen, not a general-purpose replacement for Signal.

Workplace and business messaging

In workplace and business tools the operator is not the only party who can read your messages: your employer, or the business you are talking to, controls the account, the retention and often the keys. Zoom offers the most end-to-end encryption of the five, but only when it is switched on; Teams, Slack and Zoho Cliq chat are never end-to-end encrypted; and WhatsApp Business is end-to-end encrypted only as far as the business's endpoint, which is frequently a Meta-run server.

Property	Microsoft Teams Chat	Slack	Zoom Team Chat	WhatsApp Business	Zoho Cliq
Operator	Microsoft Corporation (USA); EU Data Boundary for EU tenants	Slack Technologies, a Salesforce company (USA); data residency regions for paid plans	Zoom Communications (USA)	Meta Platforms (USA); WhatsApp Ireland for EU; plus the business and its solution provider	Zoho Corporation (India, privately held) with regional entities; Zoho-owned data centres in the region the customer chooses
Chat messages E2EE	No — chats and channel posts use TLS in transit and Microsoft 365 encryption at rest (Microsoft Learn)	No — encrypted in transit and at rest only; Enterprise Key Management lets Enterprise customers hold the master key in AWS KMS, but Slack still decrypts server-side for search, AI and exports	Optional — Advanced Chat Encryption keeps keys on participants' devices when an admin enables it; off by default	Yes to the business endpoint. Business app on a phone: full Signal-protocol E2EE. Cloud API: Meta manages the keys on the business's behalf and decrypts on Meta servers	No — Zoho states messages, calls and files are encrypted in transit and at rest; its encryption whitepaper lists Cliq chat transcripts under at-rest encryption. Reseller claims of "E2EE" are not made by Zoho itself
Calls and meetings E2EE	One-to-one VoIP calls only, off by default; scheduled meetings need a Teams Premium licence and organiser opt-in; group calls never; meeting chat never (Microsoft support)	No end-to-end encryption for huddles or calls	Optional E2EE for meetings since 2020 and Zoom Phone since 2022; post-quantum E2EE (Kyber768) for meetings since May 2024	WhatsApp voice and video calls are E2EE as on the consumer app	Calls encrypted in transit and at rest; no end-to-end encryption documented
Who else can read messages	Your organisation's Microsoft 365 admins and compliance tools; Microsoft as processor	Workspace owners and admins through exports and compliance tools (scope depends on plan); Slack as processor	Account owner and admins unless Advanced Chat Encryption is on; Zoom as processor	The business, its agents, its CRM or solution provider, and Meta's Cloud API while processing; messages retained by Cloud API for up to 30 days	Organisation admins, retention and e-discovery tools; DLP inspects message content server-side; Zoho says staff cannot view data without customer permission
Ads and data use	No ads on work data under Microsoft enterprise terms	No ads; customer messages and files feed Slack's non-generative "global models" (channel and emoji suggestions, search) unless the workspace owner emails an opt-out; Slack says generative Slack AI is not trained on customer data	No ads in paid accounts	Meta says Cloud API messages are not used to inform ads; business chats may be handled by Meta's AI agents	Ad-free; Zoho does not run an advertising business
Openness	Closed source; encryption documented	Closed source; EKM design documented	Closed source; E2EE design documented by Zoom	Closed source; WhatsApp encryption whitepaper	Closed source; encryption whitepaper published

Property	Microsoft Teams Chat	Slack	Zoom Team Chat	WhatsApp Business	Zoho Cliq
Jurisdiction and CLOUD Act	USA; CLOUD Act applies to Microsoft data wherever stored	USA (Salesforce); CLOUD Act applies regardless of data-residency region	USA; CLOUD Act applies	USA (Meta); CLOUD Act applies; plus the jurisdiction of each business and provider	Indian parent; data-residency choice by region; CLOUD Act exposure mainly for data held by Zoho's US entity

Microsoft Teams Chat

Teams chat is built for corporate record-keeping, not for privacy from the operator or the employer. Messages are encrypted in transit and at rest, and customer-managed keys are available, but they are readable by Microsoft's service and by the organisation's compliance tooling. End-to-end encryption exists only for one-to-one calls, off by default, and for Premium meetings that an organiser explicitly configures; even then the meeting chat, reactions and Q&A are not end-to-end encrypted. Treat Teams as a channel your employer can audit.

Zoom Team Chat

Zoom has the stronger encryption story of the workplace tools. Meetings can be end-to-end encrypted with post-quantum Kyber768 key exchange, and Team Chat offers Advanced Chat Encryption with device-held keys. Both are optional, and turning them on removes features such as cloud recording, continuous chat and some integrations, so most organisations leave them off. Forensic researchers have also shown that local Team Chat databases on Windows can be decrypted from the device, so endpoint security still matters (Cybersecurity News).

WhatsApp Business

When you message a company on WhatsApp, the encryption is real but its far end is the business, not a person. A small business using the WhatsApp Business app on a phone gets the same end-to-end encryption as the consumer app. A larger business using the Cloud API hands its keys to Meta, which decrypts messages on Meta servers on the business's behalf and holds them for up to 30 days; other solution providers do the same on their own servers. Meta states those messages are not used to inform ads, but they are visible to the business's agents, CRM and any AI assistant it deploys.

Slack

Slack has no end-to-end encryption in any feature: channels, DMs, huddles and files are encrypted in transit and at rest, and Slack decrypts them on its servers to power search, integrations, compliance exports and Slack AI. Enterprise Key Management moves the master key into the customer's own AWS account, which lets an organisation revoke Slack's access, but it is envelope encryption rather than E2EE. In May 2024 users discovered that Slack's privacy principles allowed customer messages and files to train its non-generative "global models" unless a workspace owner emailed an opt-out; Slack clarified that generative Slack AI runs on models hosted in its own AWS environment and is not trained on customer data (The Register). Individual employees cannot opt out themselves, and the workspace admin can export much of what they write, so Slack belongs in the same "employer can read it" category as Teams.

Zoho Cliq

Zoho Cliq is the most privacy-respecting business model among the workplace tools, but not the most private chat. Zoho runs no advertising business, hosts customer data in its own data centres in the region the customer selects, and states that staff cannot view data without permission. Messages, calls

and files are encrypted in transit and at rest only, though, and features such as data-loss prevention and e-discovery work precisely because Cliq can read message content on the server. Several Zoho resellers market Cliq as "end-to-end encrypted"; Zoho's own privacy and encryption documentation makes no such claim, so treat it as server-readable. Its Indian headquarters and regional hosting reduce, but do not eliminate, US CLOUD Act exposure.

Voice and video calls across all services

Calls are encrypted end-to-end by default on more services than text is: Signal, WhatsApp, WhatsApp Business, MEGA, Telegram, Viber, Arattai, Messenger and Discord all protect one-to-one calls without any setting, while Teams, Zoom, Google Meet and Zoho Meeting make it optional and Slack, Zoho Cliq's in-chat calls, X, LinkedIn and Instagram do not offer it.

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Service	One-to-one calls	Group calls and meetings	Notes
Signal	E2EE by default	E2EE by default (up to 50)	Same protocol stack as messages; no server access to media
WhatsApp	E2EE by default	E2EE by default	Calls relayed through Meta servers that cannot decrypt; call metadata retained
WhatsApp Business	E2EE by default	E2EE by default	Same as consumer WhatsApp for calls
MEGA Chat	E2EE by default	E2EE meetings by default	Up to 100 participants, guests can join by link
Telegram (cloud or secret)	E2EE by default, with emoji key verification	Group video chats and livestreams are not E2EE	Calls are E2EE regardless of chat type (Telegram docs)
Viber	E2EE by default	Encrypted in transit only	Group calls are the exception in Viber's otherwise default E2EE
Arattai	E2EE (encrypted before text was)	Not documented	Protocol undisclosed
Facebook Messenger	E2EE by default for personal calls	E2EE where the group thread is E2EE	Business, Marketplace and Community calls excluded
Discord	E2EE by default (DAVE)	E2EE voice channels, group DMs and Go Live	Stage channels excluded; audited by Trail of Bits
Botim	Claimed E2EE	Not documented	No protocol or audit published

Service	One-to-one calls	Group calls and meetings	Notes
Zoom	Optional E2EE with post-quantum Kyber768	Optional; disables cloud recording and some features	Off by default; all participants need supported clients
Microsoft Teams	Optional E2EE for 1:1 VoIP calls, off by default	Only Premium meetings configured by the organiser; group calls never	Meeting chat, reactions and Q&A are never E2EE
Google Chat	Hands off to Google Meet	Hands off to Google Meet	No calling of its own
Google Meet	Transit and at rest by default; optional E2EE in the consumer Meet app	Client-side encryption for Workspace Enterprise Plus organisers, external guests allowed since 2024; otherwise transit only	Recordings, in-call chat, polls and Q&A are not E2EE; dial-in unsupported in encrypted calls
Steam	Described as E2EE peer-to-peer	Same claim	No documentation or audit
Slack	Huddles not E2EE	Not E2EE	Transit and at rest only
Zoho Cliq	Transit and at rest only	Transit and at rest only	No E2EE documented for in-chat calls
Zoho Meeting	DTLS-SRTP by default; optional E2EE	Optional E2EE, enabled by the admin per meeting type (Zoho update)	Off by default; cloud recordings are host-accessible and not E2EE; no protocol documentation or audit
X Chat	No E2EE documented for calls	Not documented	Calls route peer-to-peer or via X relays depending on privacy setting
LinkedIn Messaging	Not E2EE (third-party meeting links)	Not applicable	—
Instagram DMs	Not E2EE since May 2026	Not E2EE	—
Reddit Chat	No calls	No calls	—
Filen Chats	No calls	No calls	—
Nostr DMs	No standard; client-specific experiments only	Not applicable	—
Google Messages (RCS)	Not applicable — calls hand off to the phone dialer or Google Meet	Not applicable	—
Telegram Cloud Chats	See Telegram row	See Telegram row	—

Two patterns stand out. Services that built encryption into their media stack from the start (Signal, WhatsApp, Discord's DAVE) protect calls for everyone with no trade-off, while the workplace suites bolt it on as an option that disables recording, transcription and dial-in, which is why most organisations never turn it on. And a call being end-to-end encrypted does not hide who called whom, for how long, or from which IP; only Signal's design minimises that metadata.

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Per-app profiles

WhatsApp (Meta)

WhatsApp gets content encryption right and almost everything else wrong for a privacy-sensitive user. Every chat, group and call is encrypted with the Signal protocol by default, key transparency lets clients audit the key directory, and encrypted backups exist for those who turn them on. Against that, Meta sees the full social graph, timestamps, device data and approximate location, shares account and device metadata with its other companies, and from 2025 sells ads in the Updates tab using that data. The app is also the most-targeted messenger by mercenary spyware: Paragon's Graphite in early 2025 and the CVE-2025-55177 chain in mid-2025 both achieved zero-click compromise of fully patched devices. The 2026 lockdown-style Strict Account Settings (block unknown media, silence unknown callers, disable link previews, force two-step verification) is a sensible response and the single most useful setting for high-risk users. The insider-access allegations remain unproven, but the Baig lawsuit's claim of broad engineer access to metadata is consistent with how large platforms operate and is the more credible of the two legal threads.

Facebook Messenger (Meta)

Messenger has closed most of its encryption gap with WhatsApp since December 2023: personal one-to-one chats and calls now use the Signal protocol by default, history is stored end-to-end encrypted through Meta's Labyrinth protocol, and key transparency arrived in November 2025. The gaps are the ones that matter for privacy rather than confidentiality. Messenger is tied to a Facebook account with a real-name policy, so every conversation is linked to a profile, a friend graph and an advertising identity that Meta monetises. Marketplace, business and Community Chats sit outside E2EE, and so does history from before a thread was upgraded, which means users must check each conversation rather than trust a global setting. The code is closed, there is no published independent audit, and there is no post-quantum work. Secure Storage is well designed, but users who pick Apple or Google account recovery extend their trust boundary to that account. For a privacy-sensitive user Messenger is an acceptable channel for low-stakes personal chats that are marked end-to-end encrypted, and the weakest of the Meta options for anything that reveals who you talk to.

Telegram cloud chats

The default Telegram chat is a server-side encrypted messenger, comparable in confidentiality to Slack or Discord rather than to Signal. Telegram can read, search and hand over content; it says it limits disclosures to IP and phone number, but nothing technical prevents more. The architecture is a deliberate product choice that buys instant multi-device sync, unlimited cloud history and large public channels. Since September 2024 the company has cooperated with judicial requests for ordinary crimes and the fulfilled-request count rose by an order of magnitude within months. Open-source, reproducible clients and a Dubai base outside the US and EU are the genuine positives; neither changes what the server can do.

Telegram secret chats

Secret chats add device-to-device MTProto 2.0 encryption with periodic re-keying, self-destruct timers and screenshot notifications. They cannot be used for groups, on desktop, or across a user's own devices, and they are off by default and three taps deep, so adoption is low. The MTProto design is non-standard and was found by Royal Holloway researchers to have several theoretical flaws, since fixed; it has not received the depth of formal analysis the Signal protocol has. Account metadata and every non-secret chat remain on Telegram's servers regardless.

Signal

Signal is the reference implementation of private messaging: E2EE everywhere by default, post-quantum key agreement since 2023 and a post-quantum ratchet rolling out since October 2025, Sealed Sender and private contact discovery to blind the server, open client and server code, non-profit ownership and a disclosure record that consists of account creation and last-connection dates. Its weak points are real but narrow: a phone number is still required to register, Secure Backups use a static key, and the March 2026 integrity attacks showed that features added after the core protocol (usernames, Sealed Sender) had escaped formal analysis for years. Its security depends on the host OS, as the iOS notification-storage incident showed; the fix there was Apple's. Signal's funding is donation-based, which is a sustainability question rather than a privacy one.

MEGA Chat

MEGA Chat is end-to-end encrypted by default, requires only an email address, carries no advertising business, and is built into a product whose source is public. Its cryptography is a 2015 sender-key design: static x25519 chat keys wrap per-conversation sender keys, and only membership changes or an explicit key rotation produce new keys, so it offers neither per-message forward secrecy nor post-quantum protection. The deeper issue is that chat keys hang off the same password-derived master key that ETH Zurich showed could be attacked by a malicious server in 2022 and again in 2023. MEGA patched both rounds, but no independent re-audit has been published since. Operationally MEGA discloses subscriber information routinely under New Zealand process and may consider foreign requests, and its March 2026 transparency report records a corporate restructuring that moved the contractual provider to a Hungarian entity in March 2025, a jurisdiction change users should weigh. For file-centric collaboration where encrypted storage and chat live together, it is a reasonable choice; as a pure messenger it trails Signal clearly.

Scorecard

Signal leads on every criterion except the shared weakness of needing a phone number; every service without default end-to-end encryption for text totals 3.4 or less, and Instagram DMs finish last at 1.7. Each cell is a 0–10 score, rows are sorted by the weighted total, and the weights come from the methodology section.

Service	Content (20%)	Metadata (15%)	Sharing (10%)	Legal (10%)	Openness (15%)	Backups (10%)	Track record (10%)	Defaults (10%)	Total (/10)
Signal	10	9	10	9	10	8	8	9	**9.3**
Nostr DMs (NIP-17)	5	7	9	8	9	6	5	4	**6.6**
Filen Chats	5	5	8	7	6	5	5	6	**5.8**
MEGA Chat	6	5	7	6	6	5	4	6	**5.7**
WhatsApp	8	3	2	4	4	6	4	7	**5.0**
Telegram Secret	6	3	6	5	5	7	4	3	**4.9**
Viber	7	3	3	5	3	4	5	7	**4.7**
Arattai	6	3	7	5	2	4	5	6	**4.7**
Facebook Messenger	7	2	1	3	3	7	5	6	**4.4**

Service	Content (20%)	Metadata (15%)	Sharing (10%)	Legal (10%)	Openness (15%)	Backups (10%)	Track record (10%)	Defaults (10%)	Total (/10)
Google Messages (RCS)	6	3	3	4	4	4	5	5	**4.4**
Zoom Team Chat	4	3	5	4	3	4	5	3	**3.8**
WhatsApp Business	4	3	3	4	3	3	4	5	**3.6**
Telegram Cloud	2	3	6	4	5	2	4	2	**3.4**
Zoho Cliq	1	3	7	5	2	3	5	2	**3.2**
Botim	5	2	3	2	1	3	4	5	**3.2**
Discord	3	3	4	3	4	2	2	3	**3.1**
Microsoft Teams Chat	2	3	5	4	2	3	4	3	**3.1**
Steam Chat	2	3	6	4	1	2	6	2	**3.0**
Google Chat	1	3	4	4	2	3	5	2	**2.8**
Slack	1	3	4	4	2	3	4	2	**2.7**
X Chat	3	2	2	3	1	3	3	3	**2.5**
Reddit Chat	1	4	3	4	1	2	5	1	**2.5**
LinkedIn Messaging	1	2	2	3	1	2	5	1	**2.0**
Instagram DMs	1	2	1	3	2	2	2	1	**1.7**

Scoring notes, by the cells most likely to be disputed:

- WhatsApp content encryption is 8 not 10 because the implementation is closed and has no post-quantum roadmap; it would fall further only if the insider-access allegations were substantiated.
- Telegram secret chats get 6 for encryption: real E2EE with re-keying, but a bespoke protocol with thinner analysis, no groups and no post-quantum work.
- Signal metadata is 9 not 10 because a phone number is still mandatory at registration, even though usernames hide it from other users.
- MEGA's 6 for encryption reflects default E2EE offset by the lack of a per-message ratchet; its 4 for track record reflects the 2022–23 breaks and the absence of a published re-audit, not any 2024–26 incident.
- Telegram's 6 for monetisation (above WhatsApp's 2) is because its ads live in public channels and are not fed by a cross-company profile; it would drop if Telegram Ads began using private-chat metadata.
- Signal's track record is 8 not 10 because the Sealed Sender injection flaw had existed since 2018 before researchers found it, even though the fix took eight days.

Recommendations and hardening

Use Signal for anything you would not want the operator, an advertiser or a court order to see; treat WhatsApp as encrypted-in-transit but surveilled-in-aggregate; treat Telegram cloud chats as public to Telegram.

Use case	Recommended	Acceptable fallback	Avoid
Sensitive personal or business conversations	Signal	MEGA Chat (if the other party is already on MEGA)	Telegram cloud chats; Facebook Messenger; any social or gaming platform DM
Journalists, activists, high-risk individuals	Signal with usernames, Registration Lock and notification content hidden	WhatsApp with Strict Account Settings on, for contacts who refuse to move	Telegram in any mode (phone number and IP disclosable); Facebook Messenger (identity and social graph exposed); Instagram, LinkedIn, Reddit, Steam and Discord text
Everyday chat with contacts who will not switch	WhatsApp with E2EE backups enabled	Telegram secret chats for the few 1:1 threads that matter	Default WhatsApp backup; Messenger Marketplace or business threads
Professional networking and recruiting	First contact on LinkedIn, then move anything confidential (salary, offers, documents) to Signal or email with encryption	WhatsApp	Sending contracts, IDs or compensation details in LinkedIn messages
Gaming and community groups	Discord voice channels (E2EE by default)	Steam voice	Sharing personal details, IDs or credentials in Discord or Steam text
Social contacts met on Instagram or Reddit	Move to Signal or WhatsApp once trust is established	Facebook Messenger 1:1 chat marked E2EE	Anything sensitive in Instagram DMs or Reddit Chat
Work chat with colleagues	Your employer's approved tool, assuming the employer can read it; Zoom with Advanced Chat Encryption where allowed	Teams 1:1 calls with E2EE switched on	Personal, medical or job-search conversations in Teams, Slack, Zoho Cliq or Zoom chat
Talking to companies on WhatsApp	Share only what the business needs; assume its agents, CRM and Meta's Cloud API can read it	Company's own encrypted portal for documents	Sending ID documents, card numbers or passwords to a business chat
Calling and messaging where VoIP is restricted	Signal or WhatsApp where they work	Botim or Viber for calls that must connect, keeping sensitive content for Signal	Treating a regulator-licensed app as equivalent to audited E2EE
Replacing WhatsApp for family groups	Signal	Viber or Arattai (default E2EE, closed protocol)	X Chat, Google Chat, Instagram
No company in the loop	Nostr DMs with a NIP-17 client and a remote signer	Signal (phone number required)	NIP-04 legacy Nostr DMs
Default texting on Android	Google Messages with RCS encryption on and SMS fallback off for sensitive contacts	Signal	Plain SMS/MMS
Large public communities and broadcast	Telegram channels (content is public anyway)	WhatsApp Channels	Expecting any privacy here
Encrypted file exchange plus chat in one place	MEGA Chat or Filen Chats (German jurisdiction)	Signal with attachments	WhatsApp (media may land in unencrypted backups)
Compliance with no-cloud-copy policies	Signal without backups, or Telegram secret chats	Signal with Secure Backups	WhatsApp or Telegram cloud

Hardening steps that change the outcome, per app:

1. Signal: enable Registration Lock; set a username and hide your number from everyone; set notification content to Name Only or hidden; set default disappearing-message timers; keep Secure Backups off unless you need recovery, and store the 64-character key offline if you turn them on.
2. WhatsApp: turn on end-to-end encrypted backups, two-step verification and Strict Account Settings (Settings > Privacy > Advanced); use Advanced Chat Privacy in sensitive groups; disable Meta AI where the region allows; remember the counterpart's backup is outside your control.
3. Facebook Messenger: confirm each sensitive thread shows the end-to-end encryption notice; turn on Secure Storage with a PIN or 40-character code stored offline rather than Apple or Google recovery; enable two-factor authentication on the Facebook account and review logged-in devices; keep Marketplace and business conversations free of anything sensitive; leave Meta AI out of private chats.
4. Telegram: set a two-step verification cloud password (SIM-swap defence); switch Phone Number visibility to Nobody; use secret chats with a self-destruct timer for anything sensitive; assume everything else is readable by Telegram and disclosable to authorities.
5. MEGA Chat: use a long unique password and store the recovery key offline, because the password is the master key; enable key rotation on any group that should exclude past or future members; verify contacts' identity-key fingerprints; use two-factor authentication on the account.
6. Social and gaming DMs (Discord, Steam, Reddit, LinkedIn, Instagram): treat every text message as readable by the operator; restrict who can message you to friends or connections; turn on two-factor authentication; on LinkedIn switch off Data for Generative AI Improvement and ad-sharing settings; avoid uploading government ID for age checks unless required, and delete DMs you no longer need.
7. Teams, Slack, Zoho Cliq, Zoom and WhatsApp Business: switch on E2EE for Teams 1:1 calls and Zoom meetings where policy allows; ask admins about Zoom Advanced Chat Encryption and whether the Slack global-model opt-out has been sent; keep personal matters off work accounts; when messaging a business on WhatsApp, check its business profile is verified and never send passwords or full card numbers.
8. Viber, Arattai, Botim, X Chat and Google Chat: verify encryption keys with contacts where the app offers it (Viber, Botim, Arattai); on Viber hide your number from groups you join and set a PIN; on X Chat use a long PIN and assume X can recover keys; keep anything sensitive out of Google Chat; and treat super-app wallets as identity data.
9. Nostr and Google Messages: on Nostr use a client that sends NIP-17 gift-wrapped DMs, keep your private key in a remote signer (NIP-46) or hardware, and connect to relays over Tor or a VPN; on Google Messages check for the lock icon on each chat, turn off SMS fallback for sensitive contacts, and remember iPhone contacts are only encrypted once both sides have the MLS update.
10. All apps: keep the OS patched — the two most consequential 2025–26 exposures (WhatsApp CVE-2025-55177 chain, iOS CVE-2026-28950) were OS-level or OS-chained and bypassed the messenger's encryption entirely.

Sources

All sources were consulted on 2 October 2026. Vendor documentation is listed first, then court and research material, then press.

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Signal: Introducing Secure Backups	Vendor blog	Backup design, recovery key, tiers
Signal: Government requests	Vendor transparency page	Data Signal can disclose

Source	Type	Used for
Signal: Sealed Sender and Private Contact Discovery	Vendor blog	Metadata protections
WhatsApp Security Whitepaper and E2EE backups FAQ	Vendor documentation	Protocol, backup model
Meta Engineering: WhatsApp key transparency	Vendor blog	Key verification
Telegram: MTProto and End-to-End Encryption (secret chats)	Vendor documentation	Cloud vs secret chat cryptography
Telegram transparency channel	Vendor transparency	Disclosure policy wording
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